

Linux Fundamental Notes

Working with the Shell

What is Shell?

A Linux shell is a command-line interpreter, a program that accepts commands you type, interprets them, and tells the Linux operating system what to do.

Note: Every User has a unique home directory

Home Directory

A system created directory that contains the home directory for almost all users in the linux system.

Allows Users to store their personal data in a form of file in directories.

tilde (~) represents the /home directory in the command line.

Note: The name of the user's home directory is by default identical to that of the name of the user. (e.g., /home/shem)

Command and Argument

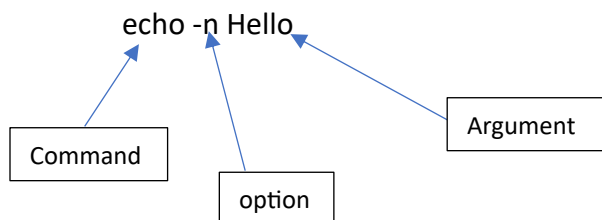
What is Command and Argument

A Linux command is an instruction you give to the shell to perform a specific task. An argument is additional information that tells the command what to operate on or how to operate.

Basic Syntax

`command [options] [arguments]`

example:



Command Types

Internal (Built-in Commands)

built directly into the shell (such as Bash). They are executed by the shell itself and do not require a separate executable file.

External Command

Linux external commands are commands that exist as separate executable files on the filesystem. When you run one, the shell searches for the executable (using the PATH environment variable), loads it into memory, and executes it.

Basic Linux Command

Command	Use
pwd	Display the current working directory.
ls	List files and directories.
cd	Change the current directory.
mkdir	Create a new directory.
rmdir	Remove an empty directory.
rm	Delete files or directories.
cp	Copy files or directories.
mv	Move or rename files and directories.
touch	Create an empty file or update a file's timestamp.
cat	Display or concatenate file contents.
less	View file contents one page at a time.
head	Display the first lines of a file.
tail	Display the last lines of a file.
echo	Display text or variable values.
clear	Clear the terminal screen.

Command	Use
history	Show previously executed commands.
man	Display the manual page for a command.
help	Display help for shell built-in commands.
which	Show the path of an executable command.
type	Determine whether a command is built-in or external.
find	Search for files and directories.
grep	Search for text patterns in files.
chmod	Change file or directory permissions.
chown	Change file or directory ownership.
ps	Display running processes.
top	Monitor running processes and system resources.
kill	Terminate a running process.
df	Display disk space usage.
du	Display file or directory disk usage.
free	Show memory usage.
uname	Display system information.
hostname	Display or set the system hostname.
whoami	Display the current logged-in user.
id	Display user and group IDs.
sudo	Execute a command with superuser privileges.
apt	Manage packages on Debian/Ubuntu systems.
yum	Manage packages on Red Hat/CentOS systems.

Command	Use
dnf	Modern package manager for Fedora/RHEL.
systemctl	Manage system services and the system state.
service	Start, stop, or check services (legacy command).
ping	Test network connectivity to a host.
ssh	Connect securely to a remote system.
scp	Securely copy files between systems.
curl	Transfer data to or from a server.
wget	Download files from the internet.
tar	Archive or extract files.
zip	Compress files into a ZIP archive.
unzip	Extract ZIP archives.

Tip: Run `compgen -c` to display all commands available in the current shell (including built-ins, aliases, functions, and external commands). Run `ls /usr/bin` to list many of the executable programs installed on your system.

Command Line Help

Command Line Help refers to the built-in documentation and help utilities available in Linux that allow users to learn how commands work, view their syntax, and understand the available options and arguments.

Command	Use
<code>man <command></code>	Displays the complete manual page for a command.
<code><command> --help</code>	Shows a quick summary of the command's syntax and options.
<code>help <command></code>	Displays help for Bash built-in commands.

Command	Use
info <command>	Displays detailed GNU documentation (if available).
whatis <command>	Displays a one-line description of a command.
apropos <keyword>	Searches for commands related to a keyword.

Shell Type

Linux shell types are the different command-line interpreters available in Linux. Each shell provides a way for users to interact with the operating system, but they differ in features, scripting capabilities, syntax, and user experience.

Shell	Description	Common Executable
Bash (Bourne Again Shell)	The default shell on most Linux distributions. Popular for scripting and system administration.	bash
Sh (Bourne Shell)	The original Unix shell. Used for compatibility and shell scripting.	sh
Zsh (Z Shell)	An enhanced shell with powerful auto-completion, themes, and customization.	zsh
Ksh (Korn Shell)	An advanced shell combining features of the Bourne and C shells. Common in enterprise Unix systems.	ksh
Csh (C Shell)	Uses syntax similar to the C programming language. Mainly found on BSD and older Unix systems.	csh

Note: run echo \$SHELL command to know what shell your system is using

Bash Shell Features

Bash (Bourne Again Shell) is the most widely used Linux shell. It provides many features that make working with Linux easier, including command execution, scripting, automation, and command-line editing.

Feature	Description
Command Execution	Executes Linux commands entered by the user.
Command History	Stores previously executed commands, allowing you to reuse them.

Feature	Description
Tab Completion	Automatically completes command names, file names, and directory names when you press <code>Tab</code> .
Command-Line Editing	Lets you edit commands using the arrow keys and keyboard shortcuts before executing them.
Aliases	Creates shortcuts for frequently used commands.
Shell Variables	Stores values that can be used by the shell or scripts.
Environment Variables	Shares variables with child processes and programs.
Shell Scripting	Automates repetitive tasks using Bash scripts (<code>.sh</code>).
Job Control	Runs, pauses, resumes, and manages background and foreground jobs.
I/O Redirection	Redirects input and output to or from files.
Pipelines	Passes the output of one command as input to another command using <code>`</code>

Bash Environment Variable

Stores information about the user's login session which is used by the shell when executing commands.

Path Variable

The PATH variable is a special environment variable that tells the Linux shell where to look for executable commands.

When you type a command like `ls`, `cp`, or `mkdir`, the shell searches the directories listed in PATH until it finds the executable file.

Bash Prompt

The Bash prompt is the text displayed by the Bash shell that indicates it is ready to accept a command from the user. Provide meaningful information about the logged-in user, name of the system, and the location on the system.

Linux Code Concepts

Linux Kernel

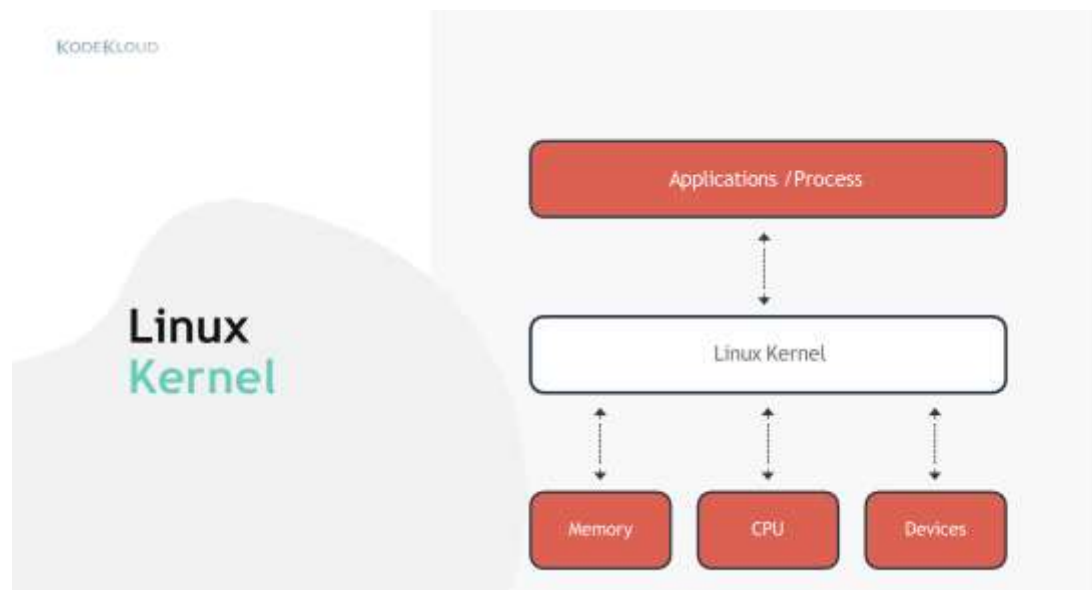
The Linux kernel is the core (heart) of the Linux operating system. It acts as a bridge between applications (including the shell) and the computer's hardware. The kernel manages system resources such as the CPU, memory, storage devices, and hardware peripherals, ensuring that programs can use them safely and efficiently.

Monolithic – carries out CPU scheduling, memory management, and several other operations by itself.

Modular – Can extend its capabilities through the use of dynamically loaded kernel modules

Kernel

Major component of an operating system and is the core interface between a computer hardware and its processes.

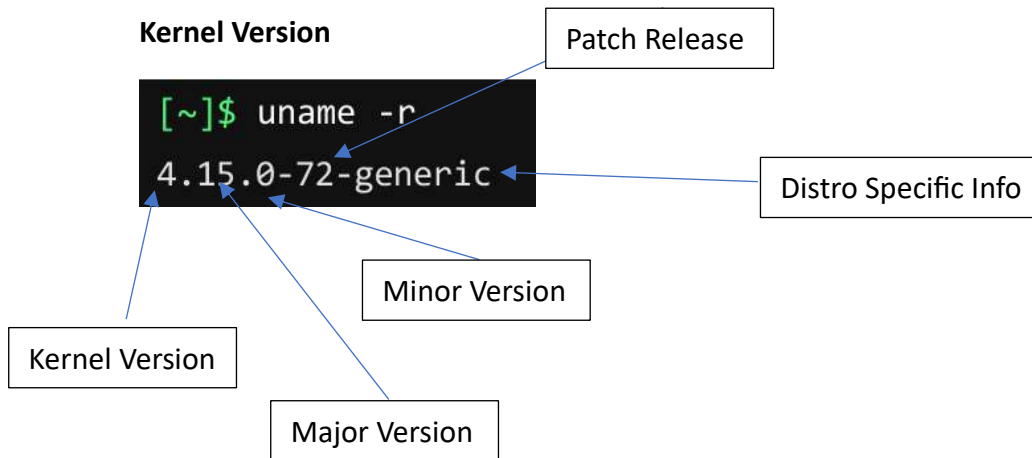


Kernel 4 Major Tasks

1. **Memory Management** – Keep track of how much memory is used to store what and where
2. **Process management** – determine which process can use the CPU, when, and for how long
3. **Device Drivers** – act as a mediator or an interpreter between the hardware and processes

4. System calls and security – Received requests for services from the processes

Tips: run `$uname` to display information about the kernel



Linux Kernel Memory Management

Kernel Space – The portion of memory in which the kernel executes and provides its services

(Process Running in Kernel Space has unrestricted access to the interface, as such it is strictly reserved for running the kernel code, kernel extension, and most device drivers)

User Space - User Space is the part of the Linux operating system where user applications and programs run. Programs in user space cannot directly access hardware. Instead, they must request services from the Linux kernel through system calls.

(All User Program get access to data by making special requests to the Kernel called system calls)

Working with the Hardware

udev services – Responsible for dynamically creating a device node associated with the newly attached drive in the slash dev file system.

dmesg – used to display messages from an area of the kernel called the ring buffer

udevadm utility – A management tool for udev. Queries the udev database for device info

example:

1. Udevadm info --query=path --name=/dev/sda5
Querying the hard disk attached to the system
2. Udevadm monitor
Listen to the kernel uevent upon detecting an event it prints the details such as path and device name.

PCI – Peripheral Component Interconnect

lspci – list PCI, display information about all PCI devices that are configured in the system

lsblk – list information about the block devices

lscpu – provides details information such as cpu architecture, number of cores, threads, model, etc.

lsmem – list available memory in the system

free -m – see information about the memory. Shows the total versus used memory in the system. (-m option used to print result in MB, -k for KB and -g for GB)

lshw – to extract detailed information of the entire hardware configuration of the machine. Can report exact memory configuration, firmware version, mainboard configuration, CPU version and speed, cache configuration, bus speed, etc.

Linux Boot

1. Bios Post – POST (Power-On Self-Test)
 - Bios run post test to ensure that the hardware components attached to the device are functioning correctly.
2. Boot Loader – Provides the user with the boot screen, often with multiple option to boot into
3. Kernel Initialization – GRUB2 (Ground Unified Bootloader Version 2)
 - Carries out task such as initializing hardware and memory management task, among other things
4. Init Process – Sets up the user space and the processes needed for the user environment

systemd

systemd is the system and service manager used by most modern Linux distributions. It is the first process started by the Linux kernel during boot (Process ID 1) and is responsible for starting, stopping, and managing system services.

systemd Targets

A systemd target is a logical grouping of system services and units that defines the state or operating mode of a Linux system.

Tips: run (systemctl get-default) to see default target. Run (systemctl set-default [runlevel].target) (P.S runlevel can be multi-user or graphical)

FileTypes

In Linux, everything is treated as a file, including regular files, directories, hardware devices, and communication channels. Each file has a file type, which is shown as the first character in the output of the ls -l command.

1. Regular File - Stores user data or program files.
2. Directory - A container used to organize files and subdirectories.
3. Special File
 - Character File - Represents devices that process data one character at a time.
 - Block File - Represents storage devices that transfer data in blocks.
 - Socket - Used for **two-way communication** between processes
 - Named Pipes - A special file used for **one-way communication** between processes.
 - Links – Ways to associate two or more filename
 - Hard link – Associate two or more file names that shares the same block of data on the physical disk
 - Symbolic link – act as pointers to another file

File System Hierarchy

The Linux File System Hierarchy (FHS) is the standard directory structure used by Linux. It organizes files and directories into a single hierarchical tree that starts from the root directory (/).

DIRECTORY	PURPOSE
/	Root directory; the top of the Linux file system.
/BIN	Essential user commands (e.g., ls, cp, mv).
/BOOT	Bootloader files and the Linux kernel.
/DEV	Device files representing hardware devices.
/ETC	System-wide configuration files.
/HOME	Home directories for regular users.
/LIB	Essential shared libraries needed by system programs.
/MEDIA	Automatically mounted removable devices (USB, CD/DVD).
/MNT	Temporary mount point for filesystems.
/OPT	Optional or third-party software packages.
/PROC	Virtual filesystem containing process and kernel information.
/ROOT	Home directory of the root (administrator) user.
/RUN	Runtime data used by running processes.
/SBIN	Essential system administration commands.
/SRV	Data for services provided by the system (e.g., web server files).
/SYS	Virtual filesystem containing hardware and kernel information.
/TMP	Temporary files (often cleared after reboot).
/USR	User applications, libraries, and documentation.
/VAR	Variable data such as logs, cache, mail, and databases.

Package Management

Linux Package Management is the process of installing, updating, removing, searching, and managing software packages on a Linux system using a package manager.

Software Package – A compressed archive that contains all the files that are required by a particular software to run.

Function of Package Manager

- **Package Integrity and authenticity** - Verifying digital certificate and checksums
- **Simplified Package Management** – Provides powerful querying options, making it easier to look up packages and then download, install, or update existing software from a software repository
- **Grouping Packages** - Reduce user confusion
- **Managing Dependencies** – Ensure a package is installed with all the packages it requires, thus avoiding what is commonly known as dependency hell.

Type of Package Managers

- **dpkg** - base package manager for Debian0based distro
- **apt** – newer front-end for dpkg system found in Debian-based distribution such us ubuntu, linux mint, and elementary OS
- **apt-get** – the traditional front-end for dpkg system found in Debian-based distro
- **rpm** – based package manager found in Red Hat-based distribution such as Red Hat Enterprise Linux, Centos OS, and Fedora
- **Yum** – front-end for the rpm system, found inv Red Hat-based distro
- **Dnf** - a more feature-richh front-end for the RPM system

Red Hat Package Manager

Used ubRed Hat Enterprise Linux, Uses rpm file extention

Yum Package Manager (Yellow Dog Updater Modifier)

Free open-source package manager that works on rpm-base Linux system.

Sequence while installing package

1. Run a transaction check
2. Transaction summary review
3. Enter y to proceed with the installation

Dpkg Utility

Low level package manager

Apt/ Apt Get

Advance Package Tool, more user-friendly. Act as package manager that relies on the dpkg utility. Relies on a software repository that contains packages that would eventually be installed on the sytem

Linux Compression and Archival

Archiving is the process of combining multiple files and directories into a single file for easier storage, backup, or distribution. Archiving does not reduce file size unless compression is also applied.

du command (disk usage) – popular command to inspect the size of the file

tar – used to group multiple files a directory into a single file.

Compression is the process of reducing the size of one or more files to save disk space and reduce transfer time

COMMAND	USE
TAR	Create or extract archive files.
GZIP	Compress a file into .gz.
GUNZIP	Decompress a .gz file.
BZIP2	Compress a file into .bz2.
BUNZIP2	Decompress a .bz2 file.
XZ	Compress a file into .xz.
UNXZ	Decompress a .xz file.
ZIP	Compress files into a .zip archive.
UNZIP	Extract .zip archives.

Searching for Files and Directory

Linux provides several commands to search for files and directories based on their name, location, type, permissions, owner, size, or content.

Common Search Commands

COMMAND	USE
FIND	Search for files and directories using many criteria (name, type, size, owner, permissions, etc.).
LOCATE	Quickly search files by name using a pre-built database.
WHICH	Find the location of an executable command in the PATH.
WHEREIS	Locate a command's binary, source code, and manual page.
GREP	Search for text or patterns inside files.

I O Redirection

is a Linux feature that allows you to redirect the input or output of a command to a file, another command, or a device instead of using the keyboard or terminal screen.

Standard Input – Accept text as an Input

Standar Output – Standard output stream

Standard Error – Coming from the standard error stream

STREAM	FILE DESCRIPTOR	DEFAULT
STANDARD INPUT (STDIN)	0	Keyboard
STANDARD OUTPUT (STDOUT)	1	Terminal (Screen)
STANDARD ERROR (STDERR)	2	Terminal (Screen)

Output Redirection (>)

Redirects **standard output (stdout)** to a file.

command > file

Append Output (>>)

Appends output to the end of a file without overwriting it.

command >> file

Input Redirection (<)

Uses a file as the input for a command.

command < file

Error Redirection (2>)

Redirects **standard error (stderr)** to a file.

command 2> error.log

Redirect Output and Errors

Save both **stdout** and **stderr** to the same file.

command > output.log 2>&1

Pipes (|)

A **pipe** sends the output of one command directly to another command.

command1 | command2

Linux Text Editors

A **Linux text editor** is a program used to **create, edit, and modify text files**, configuration files, scripts, source code, and other plain text documents directly from the command line or a graphical interface.

TEXT EDITOR	TYPE	DESCRIPTION
VI	CLI	Standard text editor available on almost all Linux systems.
VIM	CLI	Enhanced version of vi with additional features.
NANO	CLI	Beginner-friendly text editor with on-screen shortcuts.
EMACS	CLI/GUI	Powerful and highly customizable text editor.
GEDIT	GUI	Default graphical text editor for GNOME desktops.

Linux Text Editor Mode

MODE	PURPOSE
COMMAND MODE	Navigate, copy, paste, delete, and save files.
INSERT MODE	Type and edit text.
LAST LINE (COMMAND-LINE) MODE	Save, quit, and execute editor commands.

Basic Commands

COMMAND	USE
I	Enter Insert mode before the cursor.
ESC	Return to Command mode.
:W	Save the file.
:Q	Quit.
:WQ	Save and quit.
:Q!	Quit without saving.
DD	Delete the current line.
YY	Copy (yank) the current line.
P	Paste below the cursor.
/TEXT	Search for text.

Linux Security and File Permissions

Linux Security protects the operating system, users, files, and processes from unauthorized access or modification.

Linux Accounts

A Linux account is a user identity that allows a person or system process to log in, access resources, and perform tasks on a Linux system.

Linux Account Types

There are **three main types of Linux accounts**.

Account Type	UID Range (Typical)	Purpose
Root Account	0	System administrator with full privileges.
Regular (User) Account	1000+	Used by people to log in and use the system.
System/Service Account	1–999 (varies by distro)	Used by services and applications, not by human users.

User Management Commands

Command	Syntax	Use
useradd	sudo useradd <username>	Create a new user account.
adduser	sudo adduser <username>	Create a new user (interactive, Debian/Ubuntu).
usermod	sudo usermod [options] <username>	Modify an existing user account.
userdel	sudo userdel <username>	Delete a user account.
userdel -r	sudo userdel -r <username>	Delete a user and their home directory.
passwd	sudo passwd <username>	Set or change a user's password.
chage	sudo chage <username>	Manage password aging and expiration.
id	id <username>	Display a user's UID, GID, and groups.

whoami	whoami	Display the current logged-in user.
who	who	Show currently logged-in users.
last	last	Show login history.

Linux Group

A group in Linux is a collection of user accounts that share the same access permissions to files, directories, and other system resources.

Instead of assigning permissions to each user individually, you can assign permissions to a group, and all members of that group inherit those permissions.

Group Management Commands

Command	Syntax	Use
groupadd	sudo groupadd <group>	Create a new group.
groupmod	sudo groupmod <group>	Modify a group.
groupdel	sudo groupdel <group>	Delete a group.
gpasswd	sudo gpasswd -a <user> <group>	Add a user to a group.
gpasswd -d	sudo gpasswd -d <user> <group>	Remove a user from a group.
groups	groups <username>	Display the groups a user belongs to.
newgrp	newgrp <group>	Switch to another group for the current session.

Permission Types

Symbol	Permission	Value	Meaning
r	Read	4	View file contents / List directory contents
w	Write	2	Modify file / Create or delete files in a directory
x	Execute	1	Run a program / Enter a directory

Permission Groups

GROUP	MEANING
OWNER (U)	User who owns the file
GROUP (G)	Members of the file's group
OTHERS (O)	Everyone else

Numeric (Octal) Permissions

NUMBER	PERMISSION
7	rwX (4+2+1)
6	rw- (4+2)
5	r-X (4+1)
4	r--
3	-wX
2	-w-
1	--X
0	---

chmod (Change Permissions)

Numeric Mode

(chmod 755 script.sh) –

Meaning:

- Owner → rwX
- Group → r-X
- Others → r-X

Symbolic Mode

(chmod u+x script.sh) - Add execute permission for the owner.

(chmod g-w file.txt) - Remove write permission from the group.

(chmod o+r file.txt) - Give others read permission.

chown (Change Owner)

Change the file owner – (sudo chown alice file.txt)

chgrp (Change Group) – (sudo chgrp developers file.txt)

Change the file group – (sudo chgrp developers file.txt)

Directory Permissions

PERMISSION	MEANING
R	List directory contents
W	Create, rename, or delete files
X	Enter (access) the directory

Special Permissions

PERMISSION	SYMBOL	NUMERIC	PURPOSE
SUID	s	4	Run a program with the file owner's privileges
SGID	s	2	Run with the group's privileges or inherit group on new files (directories)
STICKY BIT	t	1	Only the file owner can delete files in a shared directory

Useful Commands

COMMAND	USE
LS -L	View file permissions
CHMOD	Change file permissions

CHOWN	Change file owner
CHGRP	Change file group
UMASK	View or set default permissions for new files

Linux Access Control Files

Access Control Files are system configuration files that control which users or groups are allowed or denied access to specific services, commands, or system resources.

These files help improve system security by restricting access based on user or group membership.

Common Access Control Files

File	Purpose
/etc/passwd	Stores user account information.
/etc/shadow	Stores encrypted user passwords.
/etc/group	Stores group information.
/etc/gshadow	Stores secure group passwords and administrators.
/etc/sudoers	Defines which users or groups can run commands as another user (usually root).
/etc/hosts.allow	Allows specific hosts to access network services (TCP Wrappers).
/etc/hosts.deny	Denies specific hosts from accessing network services (TCP Wrappers).

SSH (Secure Shell)

SSH (Secure Shell) is a secure network protocol used to remotely connect to and manage another computer over a network.

Unlike older protocols such as Telnet, SSH encrypts all communication, including usernames, passwords, and commands, making it secure for remote administration.

SSH Components

Component	Description
SSH Client	The computer initiating the connection.
SSH Server	The remote computer accepting SSH connections.
Port 22	Default port used by SSH.
Encryption	Protects all transmitted data.

Passwordless SSH

Passwordless SSH is a method of logging into a remote Linux system without entering a password by using SSH public-key authentication.

Instead of verifying your identity with a password, the SSH server verifies that you own a private key that matches a public key stored on the server.

Note: The login is not "without authentication" it uses SSH keys instead of a password.

Linux Networking

Name Resolution

Name Resolution is the process of translating a hostname or domain name into an IP address, allowing computers to communicate over a network without users needing to remember numeric IP addresses.

Important Configuration Files

File	Purpose
/etc/hosts	Maps hostnames to IP addresses locally.
/etc/resolv.conf	Lists DNS server addresses.
/etc/nsswitch.conf	Defines the order of name resolution.
/etc/hostname	Stores the system's hostname.

DNS (Domain Name System)

DNS (Domain Name System) is a distributed naming system that translates human-readable domain names into IP addresses, allowing computers to locate and communicate with each other on a network.

Search Domain

A Search Domain is a DNS configuration that automatically appends a domain name to incomplete hostnames when resolving them.

DNS Record Types

DNS Record Types are different types of entries stored in a DNS server. Each record provides specific information about a domain, such as its IP address, mail server, or aliases.

Common DNS Record Types

Record	Full Name	Purpose
A	Address Record	Maps a domain name to an IPv4 address.
AAAA	IPv6 Address Record	Maps a domain name to an IPv6 address.
CNAME	Canonical Name	Creates an alias for another domain name.
MX	Mail Exchange	Specifies the mail server for a domain.
NS	Name Server	Identifies the authoritative DNS servers for a domain.
PTR	Pointer Record	Maps an IP address to a hostname (Reverse DNS).
TXT	Text Record	Stores text information such as SPF, DKIM, or domain verification.
SOA	Start of Authority	Contains administrative information about the DNS zone.
SRV	Service Record	Specifies the location (host and port) of a service.

nslookup (Name Server Lookup)

nslookup is a command-line utility used to query DNS (Domain Name System) servers and retrieve information about domain names and IP addresses.

It helps administrators troubleshoot DNS resolution by checking how a hostname or IP address is resolved.

Purpose of nslookup

- Resolve a domain name to an IP address.
- Perform reverse DNS lookups (IP → Hostname).
- Query specific DNS record types (A, AAAA, MX, NS, TXT, etc.).
- Troubleshoot DNS issues.

Common Options

Option	Purpose
-type=A	Query IPv4 (A) records
-type=AAAA	Query IPv6 (AAAA) records
-type=MX	Query mail exchange records
-type=NS	Query name server records
-type=TXT	Query text records
-type=PTR	Query reverse DNS (PTR) records

Switching and Routing

Switching and Routing are two fundamental networking concepts that enable devices to communicate within the same network and across different networks.

- Switching connects devices within the same Local Area Network (LAN).
- Routing connects different networks and determines the best path for data to travel.

Switching

A network switch is a device that connects computers and other devices within the same network (LAN).

It forwards data based on MAC (Media Access Control) addresses.

Characteristics

- Operates at Layer 2 (Data Link Layer) of the OSI model.
- Uses MAC addresses.
- Connects devices within the same LAN.
- Faster local communication.

Routing

A router connects different networks and forwards packets using IP addresses.

It determines the best route for data to reach its destination.

Characteristics

- Operates at Layer 3 (Network Layer) of the OSI model.
- Uses IP addresses.
- Connects different networks.
- Determines the best path using a routing table.

ip link

The ip link command is part of the ip utility (from the iproute2 package) and is used to display and manage network interfaces on a Linux system.

A network interface is the connection point that allows a computer to communicate over a network (e.g., Ethernet, Wi-Fi, or a virtual interface).

Purpose of ip link

- Display all network interfaces.
- Show the status of each interface (UP or DOWN).
- Enable or disable network interfaces.
- View MAC (hardware) addresses.
- Rename network interfaces.

Linux Routing Table

- A routing table is a set of rules used by Linux to determine where to send network packets.

- Whenever a packet is sent, Linux checks the routing table to decide the best path (route) to the destination.

Routing Table Fields

Field	Description
Destination	Destination network or host.
Gateway (via)	Router used to reach another network.
Interface (dev)	Network interface used to send packets.
Source (src)	Source IP address used for outgoing packets.
Default Route	Used when no more specific route matches.

Common Commands

Command	Purpose
ip route	Display the routing table
ip route show	Show routing entries
ip route add	Add a route
ip route del	Delete a route
route -n	Display routing table (legacy)

Trouble Shooting

1. Check Interface
2. Check DNS Resolution – nslookup
3. Check Connectivity – ping
4. Check Route – traceroute
5. Check Services
6. Check Interface

Linux Storage

Linux Storage refers to how Linux organizes, manages, and stores data on storage devices such as hard drives (HDD), solid-state drives (SSD), USB drives, and network storage.

Storage Components

Component	Description
Storage Device	Physical disk (HDD, SSD, USB).
Partition	Logical section of a storage device.
File System	Organizes and stores files on a partition.
Mount Point	Directory where a file system is attached.
Block Device	Device that stores data in fixed-size blocks (e.g., /dev/sda).

Linux Disk Partition

A disk partition is a logical division of a physical storage device (HDD or SSD). Each partition acts as an independent storage area and can have its own file system, mount point, and purpose.

Common Partition Types

Partition	Purpose
EFI System Partition (ESP)	Stores bootloader files for UEFI systems.
Root (/)	Contains the Linux operating system.
Home (/home)	Stores user files and personal data.
Swap	Used as virtual memory when RAM is full.
Boot (/boot)	Stores the Linux kernel and bootloader files.

Partition Scheme – MBR (Master Boot Record)

MBR (Master Boot Record) is the traditional partitioning scheme used to organize partitions on a storage device. It stores the boot loader and the partition table in the first sector of the disk.

It has been widely used since the early IBM PC era but has largely been replaced by GPT (GUID Partition Table) on modern systems.

Partition Scheme – GPT (GUID Partition Table)

GPT (GUID Partition Table) is the modern partitioning scheme that replaces MBR. It is part of the UEFI (Unified Extensible Firmware Interface) standard and supports large disks, many partitions, and improved reliability.

GPT is the default partition scheme on most modern Linux, Windows, and macOS systems.

Steps to Mount a New Block Device

When a new disk is attached to a Linux system, it must be partitioned, formatted, mounted, and configured for persistent mounting.

- 1. Step 1 – Detect the New Disk**

List all block devices:

```
lsblk
```

- 2. Step 2 – Create a Partition**

Start fdisk:

```
sudo fdisk /dev/sdb
```

- 3. Step 3 – Create a File System**

Format the partition.

Create an ext4 filesystem:

```
sudo mkfs.ext4 /dev/sdb1
```

- 4. Step 4 – Create a Mount Point**

Example:

```
sudo mkdir /data
```

- 5. Step 5 – Mount the Partition**

Mount it:

```
sudo mount /dev/sdb1 /data
```

- 6. Step 6 – Verify the Mount**

Using:

```
df -h
```

- 7. Step 7 – Find the UUID**

Display UUID:

```
sudo blkid
```

- 8. Step 8 – Configure Persistent Mount**

Edit:

```
sudo vi /etc/fstab
```

Linux File System

A Linux File System is the method Linux uses to organize, store, retrieve, and manage files and directories on a storage device.

Unlike Windows, which uses drive letters (C:, D:), Linux stores everything under a single directory tree that starts at the root directory (/).

Common Linux File Systems

File System	Description	Common Use
ext4	Default Linux file system	Ubuntu, Debian
XFS	High-performance journaling file system	RHEL, CentOS
Btrfs	Modern file system with snapshots	SUSE, Fedora
FAT32	Compatible with Windows/macOS/Linux	USB drives
NTFS	Windows file system	Windows partitions
exFAT	Flash storage and large USB drives	External drives

Working with ext4

ext4 (Fourth Extended File System) is the default Linux file system for many distributions such as Ubuntu and Debian. It is a journaling file system, which means it keeps a log (journal) of changes to help recover quickly after an unexpected shutdown or system crash.

Why Use ext4?

- Stable and reliable
- Fast performance
- Supports large files and partitions
- Journaling for crash recovery
- Widely supported across Linux distributions

ext4 Features

Feature	Description
Journaling	Logs file system changes for faster recovery.
Large File Support	Supports files up to 16 TB.
Large File System	Supports file systems up to 1 EB (theoretical).
Extents	Stores data more efficiently, reducing fragmentation.
Delayed Allocation	Improves performance by delaying block allocation.
Backward Compatibility	Can mount ext2 and ext3 file systems.

/etc/fstab (File System Table)

The `/etc/fstab` (File System Table) is a configuration file that tells Linux which file systems to mount automatically during boot.

Instead of manually mounting disks every time the system starts, you define them in `/etc/fstab`, and Linux mounts them automatically.

NFS (Network File System)

NFS (Network File System) is a distributed file system protocol that allows a Linux system to share directories and files over a network.

It enables a client computer to access remote files as if they were stored on its local disk.

Linux Storage

Linux Storage refers to how Linux stores, organizes, and manages data on storage devices such as HDDs, SSDs, USB drives, and network storage.

Linux storage consists of several layers that work together to make files accessible to users and applications.

DAS (Direct Attached Storage)

Direct Attached Storage (DAS) is a storage device that is directly connected to a single computer or server without using a network.

The storage can be an HDD, SSD, USB drive, or external disk connected through interfaces like SATA, SAS, NVMe, or USB.

Characteristics of DAS

Feature	Description
Connection	Directly attached to one computer
Network Required	No
Performance	High (low latency)
Sharing	Not shared by default
Cost	Low
Scalability	Limited

Advantages

- High performance (no network overhead)
- Simple to install and manage
- Low cost
- Reliable for a single system

Disadvantages

- Cannot easily share storage with multiple servers
- Limited storage expansion
- If the server fails, storage becomes inaccessible until moved or recovered

NAS (Network Attached Storage)

NAS (Network Attached Storage) is a dedicated storage device connected to a network that allows multiple computers and servers to access and share files.

Unlike DAS, NAS is accessed over a network (usually Ethernet) using file-sharing protocols such as NFS, SMB/CIFS, or AFP.

Characteristic of NAS

Feature	Description
Connection	Network (Ethernet/Wi-Fi)

Network Required	Yes
File Sharing	Yes
Multiple Clients	Supported
Cost	Medium
Scalability	Moderate

Common NAS Protocols

Protocol	Description	Common OS
NFS	Network File System	Linux/Unix
SMB/CIFS	Server Message Block	Windows
AFP	Apple Filing Protocol	macOS (legacy)

Advantages

- Centralized file storage
- Multiple users can access the same files
- Easy backups and file sharing
- Accessible from different operating systems
- Easy to expand storage

Disadvantages

- Slower than DAS because data travels over the network
- Performance depends on network speed
- Requires network configuration

SAN (Storage Area Network)

SAN (Storage Area Network) is a high-speed dedicated network that provides block-level storage to multiple servers.

Unlike NAS, which shares files, SAN shares raw storage blocks, making the storage appear as if it were a locally attached disk to the operating system.

SAN is commonly used in enterprise environments, data centers, virtualization, and database servers.

Characteristics of SAN

Feature	Description
Connection	Dedicated storage network
Network Required	Yes
Access Type	Block-level
Multiple Servers	Supported
Performance	Very High
Cost	High
Scalability	Excellent

Common SAN Protocols

Protocol	Description
Fibre Channel (FC)	High-speed dedicated SAN network
iSCSI	Sends SCSI commands over IP/Ethernet
FCoE	Fibre Channel over Ethernet

Advantages

- Extremely high performance
- Low latency
- Highly scalable
- Centralized storage management
- Ideal for databases and virtual machines

- High availability and redundancy

Disadvantages

- Expensive hardware
- Complex setup and maintenance
- Requires specialized networking knowledge

Logical Volume Manager (LVM)

LVM (Logical Volume Manager) is a storage management system in Linux that provides a flexible way to create, resize, and manage disk storage.

Instead of using fixed disk partitions directly, LVM combines one or more physical disks into a storage pool, from which logical volumes are created

LVM Components

Component	Description
PV (Physical Volume)	Physical disk or partition used by LVM
VG (Volume Group)	Pool of storage created from one or more PVs
LV (Logical Volume)	Virtual partition created from a VG
File System	ext4, XFS, etc., created on the LV

Systemd and Service Management

systemd is the default system and service manager used by most modern Linux distributions. It is responsible for booting the system, starting and stopping services, managing system targets, logging, and monitoring system processes.

Systemd Tools

systemd provides several command-line tools for managing services, logs, boot process, users, hostnames, and system configuration.

Common systemd Tools

Tool	Purpose
systemctl	Manage services, targets, and system state
journalctl	View system and service logs
hostnamectl	View or change the system hostname
timedatectl	Manage system date, time, and time zone
loginctl	Manage user login sessions
localectl	Configure system locale and keyboard layout
systemd-analyze	Analyze boot performance and startup time
bootctl	Manage UEFI boot loader (systemd-boot)